

AP CSP | Lesson 03: Variables & Sequence

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Predict each result on paper before using the programming lab. Q5 requires exactly two choices. Show the value after each relevant assignment and justify written answers.

1. An unsigned binary ticket identifier is 00011001. Which decimal integer does it represent? [1 mark]

A. 19 B. 25 C. 26 D. 49

Answer: _____

2. A program helps a buyer find a ticket order's cost. It receives a ticket count, multiplies it by the fixed price 12 and displays a total. Which statement correctly identifies the program's input and output? [1 mark]

A. Input: the fixed price 12; output: the ticket count. B. Input: the displayed total; output: the ticket count.
C. Input: the ticket count; output: the calculated total. D. Input: the program's purpose; output: multiplication.

Answer: _____

3. What is displayed by this code? [1 mark]

```
x ← 8
y ← x
x ← x + 5
DISPLAY(y)
```

A. 8 B. 13 C. 5 D. 21

Answer: _____

4. What is the final value of result? [1 mark]

```
count ← 4
result ← 6 + count * 3
```

A. 30 B. 13 C. 12 D. 18

Answer: _____

5. Consider the code below. Which TWO statements are correct? Select exactly TWO answers. [1 mark]

```
points ← 10
saved ← points
points ← points - 3
```

A. The last line tests whether points is equal to points - 3. B. The final value of points is 7.
C. The final value of saved is 10. D. The final value of saved is 7 because it always follows points.

Answer: _____

6. What is displayed? [1 mark]

```
price ← 12
total ← 3 * price
price ← 10
DISPLAY(total)
```

A. 30 B. 36 C. 10 D. 12

Answer: _____

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Written responses | Questions 7-8

7.

[4 marks]

Trace this program.

$a \leftarrow 3$

$b \leftarrow a + 5$

$a \leftarrow b * 2$

$b \leftarrow b - 1$

DISPLAY(a)

DISPLAY(b)

(a) State (a, b) after line 2 and after line 3.

(b) State the two displayed values in order.

(c) Explain why line 4 does not change a.

8.

[4 marks]

An order has 4 tickets. Each ticket costs 12, and a fee of 3 is charged once per order.

(a) Evaluate $4 * 12 + 3$.

(b) Evaluate $4 * (12 + 3)$.

(c) Choose the expression that matches the requirement and explain the difference.

(d) Using variables tickets, price and fee that already hold 4, 12 and 3, write one assignment to total for the correct calculation.

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Written responses | Questions 9-10

9.

[4 marks]

A club wants a program that receives a member's number of sessions, charges 8 per session, adds a single registration fee of 5, and displays the total. Assume a valid session count is supplied. Write a clear, ordered algorithm in everyday English. No programming syntax is required.

10.

[4 marks]

This program should charge 10 per ticket for an order of 2 tickets, but its output is wrong.

```
price ← 12
```

```
total ← 2 * price
```

```
price ← 10
```

```
DISPLAY(total)
```

(a) State the actual output and the required output.

(b) Repair the program by moving one existing line; do not add or delete a line. Write the corrected program.

(c) Explain why the change fixes the result.

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Written responses | Questions 11-12

11. [4 marks]

The user supplies 14 first and 6 second.

first $\leftarrow$ INPUT()

second $\leftarrow$ INPUT()

average $\leftarrow$ (first + second) / 2

DISPLAY(average)

(a) State first and second after both input statements.

(b) State the displayed value.

(c) What would be displayed if the calculation were changed to average $\leftarrow$ first + second / 2? Explain using the order of operations.

12. [4 marks]

A bicycle hire program receives a whole number of hours from 1 to 8. It charges 6 per hour plus a single equipment fee of 4. Assume the input is valid.

(a) Write AP pseudocode that receives hours, uses meaningful named variables for the hourly rate and fee, calculates total, and displays it.

(b) Give the expected outputs for inputs 1 and 5.
